Journal of Theoretical Biology 604 (2025) 112071

Contents lists available at ScienceDirect

Journal of Theoretical Biology

journal homepage: www.elsevier.com/locate/yjtbi

An in-depth study of the dynamics of Thornley's mathematical model in plant biology with a view to an improved model

Ati Rostami [a,b,c,*], Brodie A.J. Lawson [a,b,c], Kevin Burrage [a,b,c,d,1]

[a] School of Mathematical Sciences, Queensland University of Technology, Brisbane, Queensland, Australia
[b] The ARC Centre of Excellence for Plant Success in Nature and Agriculture, Queensland University of Technology, Brisbane, Queensland, Australia
[c] Centre for Data Science, Queensland University of Technology, Brisbane, Queensland, Australia
[d] Department of Computer Science, University of Oxford, Oxford, Oxfordshire, United Kingdom

ARTICLE INFO

Keywords:
Carbon/nitrogen allocation
Functional balance
Parameter sensitivity
Dynamical studies
Ordinary differential equations

ABSTRACT

Plants are essential for life on Earth, serving as a major source of food supply and contributing to the planet's carbon balance. Mathematical modelling is an important mechanism for predicting and optimising plant growth in agriculture, and Thornley's mathematical model (Thornley, 1997) is one of the most widely used models describing carbon and nitrogen allocation in plants. However, a formal mathematical analysis of the model's behaviour has not been performed. Our analysis of the model provides new insights into how and why the model can be inappropriate. By varying the values of some model parameters, we identify non-physical and even quite chaotic behaviour. In response, we modify Thornley's model by including additional litter terms, resulting in the elimination of these non-physical behaviours.

1. Introduction

Plants include a wide range of organisms, including vascular plants (Mauseth, 2014). Plants' cells have highly compartmentalised structures (Lunn, 2007), so cell–cell interactions with one another are intricate and not always well understood. Mathematical modelling helps to better understand biological mechanisms and is a powerful tool for developing and testing hypotheses (Küstner et al., 2019; Chew et al., 2014; Dun et al., 2009). It is also essential to understand complex plant biology phenomena (Farquhar et al., 1980). Often, these models are based around ordinary differential equations (ODEs), for example, in plant physiology (Pitchford and Avello, 2022), or to represent biochemical processes inside plants (Ban et al., 2019). ODEs are commonly used to describe the dynamics of biological systems because the processes taking place are described more naturally in terms of rates of change (Polynikis et al., 2009).

Broadly, a plant consists of two main parts, the shoot matter and the root matter. Nitrogen is absorbed from the soil through the roots, while carbon is assimilated through photosynthesis in the shoot matter (Krämer et al., 2022; Thornley, 1997). However, both the roots and the shoots need carbon and nitrogen for plant growth. To obtain these resources, the plant transports carbon and nitrogen from the shoot to the root and vice versa. This transport between the shoots and roots occurs through the phloem and xylem (Seki et al., 2015; Payvandi et al., 2014).

Lacointe (2000) reviews some of the existing models that describe the process of carbon allocation. These models include empirical allocation coefficients, transport resistance, growth rules-based models such as functional balance principles, and relative sink strength models. Many popular sophisticated crop growth models typically use phenomenological perspectives (such as empirical rules, (McCown et al., 1996)). A particularly important mathematical model that treats this in a more mechanistic way describing carbon/nitrogen allocations for plant dry matter is Thornley's model, the model considered in this study (Thornley, 1997). This model has been extended in a variety of ways, for example, by using a more complex description of photosynthesis acclimation (Thornley, 1998b), the inclusion of daily and seasonal dynamics, temperature, and light (Thornley, 1999). Thornley's initial model (Thornley, 1972) has also been used as a model for the growth of forest tree models (Sievänen et al., 2000) and, extended to a highly detailed forest growth model in identical trees of the same age (Thornley, 1991).

In particular, we highlight the work of Higgins and Richardson (2014) who states that Thornley's model is "a well-established physiological model that captures the fundamental processes that influence a plant's physiological performance". Higgins and collaborators have

* Corresponding author at: School of Mathematical Sciences, Queensland University of Technology, Brisbane, Queensland, Australia.
E-mail address: ati.rostami@hdr.qut.edu.au (A. Rostami).
[1] Visiting Professor.

https://doi.org/10.1016/j.jtbi.2025.112071
Received 28 March 2024; Received in revised form 7 February 2025; Accepted 12 February 2025
Available online 5 March 2025
0022-5193/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

used Thornley's model in a number of important studies. For example, Higgins et al. (2012) predicts where plants can grow based on how well they use resources such as carbon and nitrogen under different environmental conditions (such as temperature, moisture, and light). Higgins and Richardson (2014) considered the global introduction 749 species of Australian acacias and eucalypts by analysing environmental factors such as temperature, soil moisture, and nutrient availability. They found plant invasions are not just about specific traits but also about how well species can adapt to different environmental conditions globally. By developing a model based on Thornley's model and data collection from 100 sites spread all across the world under a number of different types of environmental conditions, Higgins et al. (2023) investigated the effects of climate change on species distribution. They found that although carbon in the atmosphere has increased, plants are less able to store carbon due to environmental factors (water and temperature), which in turn makes it harder to fight climate change. In all three of these papers, Thornley's work was key.

Dewar et al. (2002) also developed a mechanistic model for carbon and light use efficiency in plants. Moreover, a mechanistic model of tree growth and a model of phloem transport were developed by Potkay et al. (2022) to investigate the mechanics of phloem transport. Feller et al. (2015) developed a mechanistic mathematical model to describe the shoot and root activity, such as water and nutrient uptake and photosynthesis and how they interact with sugar and phosphate. One of the differences between Feller et al. (2015) and the models of Thorney (Thornley, 1998a, 1997) is that Feller considered day-night cycles. Weisberg et al. (2005) used a set of differential equations to describe dry matter allocation and their nutrient transport and conversion processes, differing from Thornley's model by including carbohydrate pools. Further models have included more than two compartments to better represent plant physiology (Linker and Johnson-Rutzke, 2005; Seginer, 2003). Dorneles et al. (2001) extended Thornleys' models of dry matter allocation (Thornley, 1998a, 1997, 1995) to a model for the case of two plants interacting as part of an environment with many species and microfauna. They realised that interactions between two plants could significantly affect the changes in their mass over time. These, together with other extensions of Thornley's model (Zaffaroni et al., 2020; Higgins and Richardson, 2014; Thornley and Parsons, 2014) demonstrate the continued interest in carbon/nitrogen allocation, and Thornley's framework for modelling this. Rohde and Forni (2023) described a mechanistic model for plant growth, particularly for lettuce, which is a simplified version of the original Thornley's model.

Despite this variety of models, there has not been a proper consideration and systematic analysis of how such models are affected by changes in their parameters. Additionally, formal mathematical analysis of these models is lacking. Such analysis can identify which parameters or conditions play an important role in controlling the overall dynamics. Furthermore, in many biological systems, parameters act in combinations, and their individual effects can be hard to determine (Monsalve-Bravo et al., 2022). For this reason, here we analyse Thornley's model in order to identify combinations of parameters that act together and hence summarise some of the model's dynamics and its steady state behaviour.

We also demonstrate how variations in some parameters of the model affect its dynamics and steady state behaviour. In particular, we observe that some parameter values result in non-physical and chaotic behaviour. After investigating the nature of these different non-physical dynamics and when they occur, we propose a correction to how the model treats losses of plant matter. This results in more reasonable outputs in these scenarios.

In Section 2, we introduce the model and provide an analysis for its steady state behaviour, hence identifying parameters that are important. We also reframe the model in terms of concentrations and hence modify the model by changing how losses of matter are represented. In Section 3, we provide a detailed description of the dynamics of carbon and nitrogen allocation in plants, as predicted by the original and modified Thornley models. By varying the parameter values, we observe unexpected behaviours in the original model, and demonstrate much more stable behaviours for the modified model over the parameter range we consider. We also demonstrate the simulations and equilibrium analysis of the modified model, and we discuss and conclude our work in Section 4.

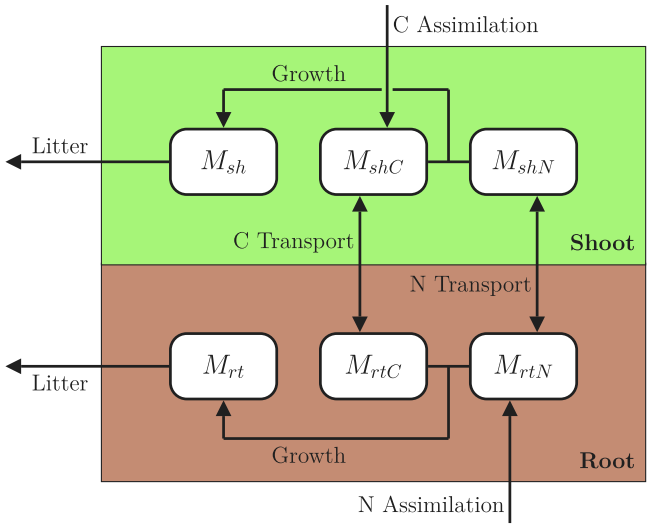


Fig. 1. The basic structure of Thornley's model of carbon and nitrogen allocation in plants. Subscripts sh and rt refer to the shoots and roots, respectively. Dry matter mass (M_{sh} and M_{rt}) is produced via growth, which requires both substrate carbon (M_{shC}, M_{rtC}) and nitrogen (M_{shN}, M_{rtN}). Substrate nutrients flow between shoot and root, with carbon produced in the shoot, and nitrogen in the roots.
Source: Picture adapted from (Thornley, 1997; Dewar, 1993).

2. Thornley's model

2.1. The model

Thornley's 1997 paper (Thornley, 1997) provides a complete description of his model of the allocation of nitrogen and carbon in the roots and shoots of plants. The model is based on transport and chemical conversion, the two major types of process in the plant that are necessary for completing allocation. The model consists of six ordinary differential equations with 13 parameters that describe these two processes, that we summarise here for completeness. Moreover, Thornley included two switch variables that adjust the behaviour between balanced exponential growth and steady state (σ_{SS}) and for turning on and off product inhibition of carbon and nitrogen acquisition (σ_{PI}). The values of these two switch variables are either zero or one and are not continuous. Fig. 1 shows a schematic diagram of the model and the different transport and conversion processes that represent its basic structure.

The model has six state variables and is given below, each denoting the mass of different components of the plant in its shoot matter and the root matter. These components are dry matter (M_{sh} and M_{rt}), substrate bound carbon (M_{shC} and M_{rtC}), and substrate bound nitrogen (M_{shN} and M_{rtN}).

$$\frac{dM_{sh}}{dt} = \underbrace{\frac{k_G M_{shC} M_{shN}}{M_{sh}}}_{\text{Growth}} - \underbrace{\frac{\sigma_{SS} k_{litt} M_{sh}}{1 + \frac{K_{M,litt}}{M_{sh}}}}_{\text{Litter Flux}}, \tag{1}$$

$$\frac{dM_{rt}}{dt} = \underbrace{\frac{k_G M_{rtC} M_{rtN}}{M_{rt}}}_{\text{Growth}} - \underbrace{\frac{\sigma_{SS} k_{litt} M_{rt}}{1 + \frac{K_{M,litt}}{M_{rt}}}}_{\text{Litter Flux}}, \tag{2}$$

$$\frac{dM_{shC}}{dt} = \underbrace{\frac{k_C M_{sh}}{\left(1+\frac{\sigma_{SS}}{K_M}M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)}}_{\text{Photosynthesis}} - \underbrace{f_C k_G \frac{M_{shC}}{M_{sh}} M_{shN}}_{\text{Utilisation for Growth}} - \underbrace{\frac{1}{x\rho_C}\left(\frac{M_{shC}}{M_{sh}} - \frac{M_{rtC}}{M_{rt}}\right)}_{\text{Transport}}, \qquad x = \left(\frac{1}{M_{sh}^q} + \frac{1}{M_{rt}^q}\right) \tag{3}$$

$$\frac{dM_{shN}}{dt} = \underbrace{\frac{1}{x\rho_N}\left(\frac{M_{rtN}}{M_{rt}} - \frac{M_{shN}}{M_{sh}}\right)}_{\text{Transport}} - \underbrace{f_N k_G \frac{M_{shN}}{M_{sh}} M_{shC}}_{\text{Utilisation for Growth}}, \tag{4}$$

$$\frac{dM_{rtC}}{dt} = \underbrace{\frac{1}{x\rho_C}\left(\frac{M_{shC}}{M_{sh}} - \frac{M_{rtC}}{M_{rt}}\right)}_{\text{Transport}} - \underbrace{f_C k_G \frac{M_{rtC}}{M_{rt}} M_{rtN}}_{\text{Utilisation for Growth}}, \tag{5}$$

$$\frac{dM_{rtN}}{dt} = \underbrace{\frac{k_N M_{rt}}{\left(1+\frac{\sigma_{SS}}{K_M}M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}\frac{M_{rtN}}{M_{rt}}\right)}}_{\text{Nitrogen Uptake}} - \underbrace{f_N k_G \frac{M_{rtN}}{M_{rt}} M_{rtC}}_{\text{Utilisation for Growth}} - \underbrace{\frac{1}{x\rho_N}\left(\frac{M_{rtN}}{M_{rt}} - \frac{M_{shN}}{M_{sh}}\right)}_{\text{Transport}}. \tag{6}$$

In Eqs. (1) and (2), positive terms represent growth rates for shoot and root, and negative terms represent litter fluxes from shoot and root. The litter term represents general loss of plant matter (for example, leaf litter in the shoot). Carbon and nitrogen are both required for growth, so Thornley (1972) assumed that the growth rate is given by the product of the two substrate concentrations. Thornley chose a form of the litter term that guarantees that a steady state is maintained (when a switch variable for steady state is turned on, $\sigma_{SS} = 1$) (Thornley, 1995).

In Eq. (3), the first positive term represents carbon assimilation from photosynthesis. If the steady state switch is on ($\sigma_{SS} = 1$), the rate of carbon assimilation decreases as the plant grows in mass, allowing the model to reach steady state. If the product inhibition switch is on ($\sigma_{PI} = 1$), then the presence of substrate carbon makes it more difficult to produce more carbon, reducing the assimilation rate. The second term represents the consumption of substrate carbon in the shoot for growth, and is proportional to the shoot's growth rate (first term in Eq. (1)). The transport of carbon from the shoot to the root is described by the third term, operating from the principle (Thornley and France, 2007; Thornley, 1995)

$$\text{Substrate Transport Rate} = \frac{\text{Concentration Difference}}{\text{Resistance}}.$$

The resistance is given by the product of the resistivity for carbon (ρ_C) and a nonlinear term (x) that depends on the dry matter mass of the plant and represents the effects of plant size and architecture on resistance.

Eq. (6) functions similarly to Eq. (3) but for nitrogen in the root, where the source term now represents nitrogen uptake through the roots instead of photosynthesis. Eqs. (4) and (5) also function similarly, except that these equations describe the amounts of nitrogen in the shoot and carbon in the root, respectively. There is no uptake term, instead these substances are transported into these compartments.

The parameters of the model, including the default values as given by Thornley, are given in Table 1. We use the value of f_N from the paper "Modelling Shoot: Root Relations: the Only Way Forward?" (Thornley, 1998a), as this appears to have been given incorrectly in the original paper (Thornley, 1997). These parameters were set by Thornley in proposing a model for generic plants, but of course, will vary for specific plants.

2.2. Steady state analysis of the original Thornley's model

In this section, we carry out a detailed analysis of the equilibria of the original Thornley model, along with a set of remarks associated with each subsection of the analysis. The equilibria can be found by setting all the time derivatives to zero. We also perform a numerical calculation/steady state eigenvalues analysis of the Jacobian in Section 3.3.

We show in Theorem 1, when the plant reaches steady state, that there is a balance between the amount of matter in the shoot and root, and this balance depends on the parameters θ and ϕ. θ is a dimensionless parameter that defines the balance between the plant's production and usage of carbon and nitrogen, and ϕ represents the extent to which nutrient uptake is limited by plant mass.

Theorem 1. *At steady state, the plant's balance of shoot/root satisfies*

$$\frac{\theta M_{sh}}{\left(1+\phi M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)} = \frac{M_{rt}}{\left(1+\phi M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}\frac{M_{rtN}}{M_{rt}}\right)}.$$

When no product inhibition ($\sigma_{PI} = 0$) is included then

$$M_{rt} = \frac{\theta M_{sh}}{1+\phi M_{sh}(1-\theta)}, \quad \text{where } \theta = \frac{k_C f_N}{k_N f_C} \text{ and } \phi = \frac{\sigma_{SS}}{K_M}. \tag{7}$$

Proof. The proof is in Appendix A. □

Eq. (7) shows the relation between M_{sh} and M_{rt} at steady state. When $\theta = 1$, so $M_{sh} = M_{rt}$, the original Thornley model predicts that the shoot and root dry matter mass of the plant are equal at steady state. In this scenario where the uptake-to-demand ratio is equal for both carbon and nitrogen, the rate of growth is equal in shoot and root, and equal also to their rates of loss due to litter.

In the scenario that the plant is dead ($M_{sh} = M_{rt} = 0$), so $M_{shN} = M_{rtC} = 0$ at steady state (see Appendix B). Following the argument in Appendix B, it can be seen that M_{shC} and M_{rtN} are non-zero in dying plants. In particular,

$$M_{shC} = \frac{f_C}{f_N} M_{rtN} + K, \qquad K \text{ is a constant.} \tag{8}$$

We show additional details in Appendix B. We observe dynamics in Section 3 that shows when the plant dies, shoot carbon and root nitrogen are still produced in large quantities. This is infeasible. Now letting $C = \frac{k_C}{\sigma_{SS} k_{litt} f_C}$, the existence of physically feasible steady states in terms of this parameter can be established in Theorem 2.

Theorem 2. *If $\theta < 1$, and $C \geq 1$, the existence of single physically feasible steady states in the original Thornley model is guaranteed.*

Proof. The proof is in Appendix A. □

Steady states in the Thornley model can be approached with or without damped oscillations. However, we also observe cases where the model oscillates around the steady state. For some parameters, the plant mass essentially approaches zero when the feasible steady state is not stable. We examine the stability of the model in more detail in Section 3.

Theorem 3. *If production (k_C and k_N) versus consumption (f_N and f_C) for carbon and nitrogen are in balance ($\theta = 1$), there is one positive real root in the original Thornley model.*

Proof. The proof is in Appendix A. □

In our analysis, the parameters θ, C, and ϕ play a significant role in determining the model's behaviour at steady state, especially θ. This analysis is consistent with the idea that some model parameters act in combinations, often expressed as ratios or products (Monsalve-Bravo et al., 2022).

Table 1
The parameters as given by Thornley (1997). Please note that the abbreviations C, N, Su, St, and dm refer to Carbon, Nitrogen, Substrate, Structural, and Dry Matter, respectively.

Symbol	Value	Unit	Definition
k_G	200	[(kg Su C)(kg Su N)(kg St dm)$^{-2}$]$^{-1}$d^{-1}	Growth rate
k_{litt}	0.05	d^{-1}	Litter rate
$K_{M,litt}$	0.5	kg St dm	Litter parameter
k_C	0.1	kg Su C (kg St dm)$^{-1}$d^{-1}	Carbon assimilation rate
k_N	0.02	kg Su N (kg St dm)$^{-1}$d^{-1}	Nitrogen uptake rate
f_C	0.5	kg Su C (kg St dm)$^{-1}$d^{-1}	Fraction of carbon in structural dry matter
f_N	0.025	kg Su N (kg St dm)$^{-1}$d^{-1}	Fraction of nitrogen in structural dry matter
J_C	0.1	kg Su C (kg St dm)$^{-1}$	Inhibition constant for carbon assimilation
J_N	0.01	kg Su N (kg St dm)$^{-1}$	Inhibition constant for nitrogen uptake
K_M	1	kg St dm	Half-saturation constant for C/N uptake
q	1		Plant architecture parameter for transport
ρ_C	1	(kg St dm)$^{q-1}$d	Transport resistance coefficient for substrate carbon
ρ_N	1	(kg St dm)$^{q-1}$d	Transport resistance coefficient for substrate nitrogen
σ_{SS}	{0,1}	(off/on)	Switch — steady state growth
σ_{PI}	{0,1}	(off/on)	Switch — product inhibition

2.3. A modified Thornley model with additional litter terms

We rewrite Thornley's model in terms of the concentrations of substrate carbon and nitrogen. These concentrations are defined as $C_{sh} = \frac{M_{shC}}{M_{sh}}$ (concentration of substrate carbon in the shoot) and $C_{rt} = \frac{M_{rtC}}{M_{rt}}$ (concentration of substrate carbon in the root) and similarly for nitrogen, $N_{sh} = \frac{M_{shN}}{M_{sh}}$ and $N_{rt} = \frac{M_{rtN}}{M_{rt}}$. These definitions of concentration match those used in deriving Thornley's model, as can be seen by looking at the transport terms that act to equalise them. Using the quotient rule for derivatives, we can write

$$\frac{dC_{sh}}{dt} = \frac{d}{dt}\left(\frac{M_{shC}}{M_{sh}}\right) = \frac{1}{M_{sh}}\frac{dM_{shC}}{dt} - \frac{M_{shC}}{M_{sh}^2}\frac{dM_{sh}}{dt}. \tag{9}$$

Considering Eqs. (1) and (3) in the original Thornley model, together with the Eq. (9), after some rearranging and writing in terms of concentrations instead of masses, we obtain

$$\begin{aligned}\frac{dC_{sh}}{dt} =& \frac{k_C}{\left(1+\frac{\sigma_{SS}}{K_M}M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}C_{sh}\right)} - f_C k_G C_{sh} N_{sh} \\ &- \frac{C_{sh}-C_{rt}}{x\rho_C M_{sh}} - k_G C_{sh}^2 N_{sh} + \frac{\sigma_{SS}k_{litt}C_{sh}M_{sh}}{M_{sh}+K_{M,litt}}.\end{aligned} \tag{10}$$

Writing the model this way, some terms appear unreasonable. Considering Eq. (10), we see that the rates of change for the carbon and nitrogen concentration has a source term (the last term in Eq. (10)). By continuing the same procedure for other concentrations, ($C_{rt} = \frac{M_{rtC}}{M_{rt}}$, $N_{sh} = \frac{M_{shN}}{M_{sh}}$ and $N_{rt} = \frac{M_{rtN}}{M_{rt}}$), we see the extra litter terms for each one. Consider that when a leaf falls off the plant, the concentrations in the remaining plant are not expected to change. Here, however, the litter process corresponds to an increase in concentration. This can also be seen in the model written in terms of masses as in Eqs. (3)–(6), where substrate carbon and nitrogen are consumed only by growth. Therefore, when the plant is not growing, it lacks a sink for substrate carbon/nitrogen and these remain stored despite its decreasing dry matter. Now we modify the model to remove the sources of concentration.

Given the unexpected source terms identified in Eq. (10) and other concentrations, we now seek a modification to the model that corrects this nonphysical behaviour. To remove these source terms, we must add a litter flux to the differential equations for substrate carbon and nitrogen also. The form of this term can be seen in the modified model,

$$\frac{dM_{sh}}{dt} = k_G C_{sh} N_{sh} M_{sh} - \frac{\sigma_{SS}k_{litt}M_{sh}^2}{M_{sh}+K_{M,litt}}, \tag{11}$$

$$\frac{dM_{rt}}{dt} = k_G C_{rt} N_{rt} M_{rt} - \frac{\sigma_{SS}k_{litt}M_{rt}^2}{M_{rt}+K_{M,litt}}, \tag{12}$$

$$\begin{aligned}\frac{dM_{shC}}{dt} =& \frac{k_C M_{sh}}{\left(1+\sigma_{SS}\frac{M_{sh}}{K_M}\right)\left(1+\sigma_{PI}\frac{C_{sh}}{J_C}\right)} - f_C k_G C_{sh} N_{sh} M_{sh} \\ &- \left(\frac{C_{sh}-C_{rt}}{x\rho_C}\right) - \underbrace{\frac{\sigma_{SS}k_{litt}M_{shC}M_{sh}}{M_{sh}+K_{M,litt}}}_{\text{Added Litter Flux}},\end{aligned} \tag{13}$$

$$\frac{dM_{shN}}{dt} = \left(\frac{N_{rt}-N_{sh}}{x\rho_N}\right) - f_N k_G C_{sh} N_{sh} M_{sh} - \underbrace{\frac{\sigma_{SS}k_{litt}M_{shN}M_{sh}}{M_{sh}+K_{M,litt}}}_{\text{Added Litter Flux}}, \tag{14}$$

$$\frac{dM_{rtC}}{dt} = \left(\frac{C_{sh}-C_{rt}}{x\rho_C}\right) - f_C k_G C_{rt} N_{rt} M_{rt} - \underbrace{\frac{\sigma_{SS}k_{litt}M_{rtC}M_{rt}}{M_{rt}+K_{M,litt}}}_{\text{Added Litter Flux}}, \tag{15}$$

$$\begin{aligned}\frac{dM_{rtN}}{dt} =& \frac{k_N M_{rt}}{\left(1+\frac{\sigma_{SS}}{K_M}M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}N_{rt}\right)} - f_N k_G C_{rt} N_{rt} M_{rt} \\ &- \left(\frac{N_{rt}-N_{sh}}{x\rho_N}\right) - \underbrace{\frac{\sigma_{SS}k_{litt}M_{rtN}M_{rt}}{M_{rt}+K_{M,litt}}}_{\text{Added Litter Flux}}.\end{aligned} \tag{16}$$

It can be seen that the same litter rates ($\sigma_{SS}k_{litt}M_{sh}/(M_{sh}+K_{M,litt})$ in shoot and $\sigma_{SS}k_{litt}M_{rt}/(M_{rt}+K_{M,litt})$) now apply to all terms in the model, such that the litter process does not cause changes in the plant's nutrient concentrations. In the original model, when a leaf falls, the plant concentrations of substrate (the availability of substrate carbon) in the plant increase but in the modified Thornley model, the litter term no longer acts to increase the concentration available in the plant.

In the case that $\sigma_{PI} = 0$ and both M_{sh} and $M_{rt} \to 0$, and let $R = \frac{M_{sh}}{M_{rt}}$ we can add (13) and (15) giving $\frac{d}{dt}(M_{shC}+M_{rtC}) = 0$ so that

$$M_{shC} + M_{rtC} = K_1, \tag{17}$$

but also from (13)

$$\frac{dM_{shC}}{dt} = -\frac{M_{shC} - M_{rtC}R}{x\rho_c M_{sh}}.$$

Substituting (17) gives

$$\frac{dM_{shC}}{dt} = -\frac{M_{shC}(1+R) - K_1 R}{x\rho_c M_{sh}},$$

which yields an exponential decay to

$$M_{shC} = K_1 \times \frac{R}{1+R}.$$

We can repeat this analysis by adding (14) and (16),
Yielding an exponential decay to

$$M_{rtN} = \frac{K_2}{1+R}.$$

Now if $\sigma_{PI} \neq 0$, as in (13) and (16), then this implies that the decay of the M_{sh} and M_{rt} solutions, as shown above, just happens more quickly than in the $\sigma_{PI} = 0$ case.

3. Dynamics of the original and the modified Thornley's model

3.1. The range of behaviours in the two models

The effects of varying parameters in the Thornley model have not been thoroughly considered, and in presenting the model Thornley (1997) considered only minor variations in their values. As the model was built using physical arguments, we expect its predictions to be physical also. If varying some of the parameters results in behaviour that is not physical, then investigating these dynamics may indicate areas where the model can be improved.

We begin with the parameters as given in Table 1, and choose the version of the model for plants that (by default) grow to a fixed size ($\sigma_{SS} = 1$). We identify q as a parameter of interest, as it appears in the model in terms on the denominator of fractions that are themselves on the denominator of the transport term, so its effects on model behaviour are difficult to predict. We also identify σ_{PI} as a modulating factor on the effects of q and allow it to vary, instead of treating it as a pure switch with value zero or one. This in effect adjusts J_C and J_N in the model together. Model simulations are generated by numerically integrating Eqs. (1)–(6) using MATLAB's ODE solver, `ode15s`. This routine uses adaptive time stepping to maintain the specified error tolerance, which we set at 10^{-6} for both the absolute and relative errors for the model species.

By exploring different values of the parameters q and σ_{PI}, we identify several interesting dynamical behaviours in the original Thornley model, that do not appear in the modified version. In Fig. 2, we demonstrate these dynamical behaviours in terms of the total mass of dry matter ($M = M_{sh} + M_{rt}$). Realistic plant behaviours include growing, reaching a fixed size, or dying, but in the original model, we see a broad range of behaviours that includes approaching steady state, potentially with damped oscillations, sustained oscillations, spiked oscillations that increase in amplitude before eventual collapse, period doubling to chaos (Alligood et al., 1996, Ch.9), and plant death. We consider these as six unique categories of behaviour that the model exhibits. The unexpected among these behaviours indicate possible deficiencies in the model, or at least regions in parameter space that result in inappropriate model output. In contrast, in our modified model, the plant either grows to steady state or fails to grow once transport becomes more sensitive to mass as q becomes large. All of the non-physical behaviours of the original model highlighted in Fig. 2 have been eliminated.

Fig. 3 shows the behaviour of the original model in greater detail, for the same set of scenarios as in Fig. 2. Considering the trajectories of all six individual species in the original model, we observe that shoot carbon and root nitrogen tend to behave the same way, even across this broad variety of dynamics. Due to the relative abundance of shoot as compared to root in the plant, at least for these parameter values, shoot carbon is present in much larger amount than root nitrogen. Importantly, the original model predicts that even when the plant dies, shoot carbon and root nitrogen are still produced in large quantities, as can be seen in the bottom left and bottom right panels of Fig. 3. We have discussed this issue in Section 2.2.

Fig. 4 shows in detail the amplitudes of the oscillations of shoot dry matter mass in the top middle graph of Fig. 3, which exhibit unexpected variation in a repetitive pattern. This occurs despite solving the ODEs to a high level of accuracy, suggesting that this is not a numerical artefact.

Fig. 5 shows the dynamics of the six model species in our modified version of Thornley's model, for the same parameter values that produce the varied dynamics in the original model shown in Fig. 3. Matching with the observations for total dry matter mass, the modified model results in much more credible model predictions with the unrealistic oscillatory dynamics observed for the original model removed. However, carbon in the shoot and nitrogen in the root remain with non-zero mass in dying plants. Specifically, we observe M_{shC}, M_{rtN}, M_{shN}, and M_{rtC} exponentially approach a stable state when the plant dies, (M_{shN} and M_{rtC} go to a very small value). This can be seen in the top right, bottom right, and bottom left panels of Fig. 5. In the scenario with $\sigma_{PI} = 0.1$, we observe that shoot carbon and root nitrogen are produced in smaller quantities and approach a small value (Fig. 5, bottom middle panel), consistent with the analysis in Section 2.3.

Looking at Fig. 5, we observe that dying plants behave more appropriately with a nonzero σ_{PI}. With product inhibition on, the amount of shoot carbon concentration (C_{sh}) in photosynthesis and root nitrogen concentration (N_{rt}) in nitrogen uptake are significantly decreased. When the plant dies, product inhibition ($\sigma_{PI} = 1$) is enough to cause these concentrations to also approach zero (Fig. 6). This was also suggested in Section 2.3, and is due to the faster movement of M_{sh} and M_{rt} towards zero. After adding extra litter terms and including product inhibition, the model's behaviour meets the goal of eliminating the most apparent non-physical behaviours. We recommend this version of the model for future studies.

Given the issue observed here, where nutrients accumulate even as the plant dies, one might consider alternatives to our litter terms to act as sinks for excess carbon or nitrogen. Indeed, neither the growth term in the original model, nor our new litter term, can effectively remove nutrient when the plant is dying and its structural mass becomes very small, as both are a process of the plant and hence their rate is set by the amount of plant dry matter. A natural consideration, then, would be litter terms that do not depend on dry matter mass. A simple way to achieve this is to introduce first-order decay terms for all of the substrate species in the model, M_{shC}, M_{rtC}, M_{shN} and M_{rtN}. These terms cannot be justified as a plant process (for example consumption of nutrient for maintenance purposes instead of growth), as they by design do not depend on how much plant is present. However, these terms could perhaps be treated as degradation of the usable forms of carbon and nitrogen. Moreover, the decay rate could likely be set small, only having a significant impact on model predictions when it becomes necessary to clear excess nutrient when the plant loses most of its mass. Given the potential of such sink terms, we return to analyse them in terms of model stability in Section 3.3.

3.2. Effects of parameter variations

In Section 3.1, we identified a range of different dynamics in the original Thornley model, but did not fully explore the parameter space of q and σ_{PI}. Here, we thoroughly explore how variations in these two parameters control the dynamics in both the original and modified models. This also serves to verify that the more reasonable carbon and nitrogen allocation dynamics we observed for the modified version of the model remain a consistent trend throughout the space of these two parameters. To do this, we first simulate the model for a long period

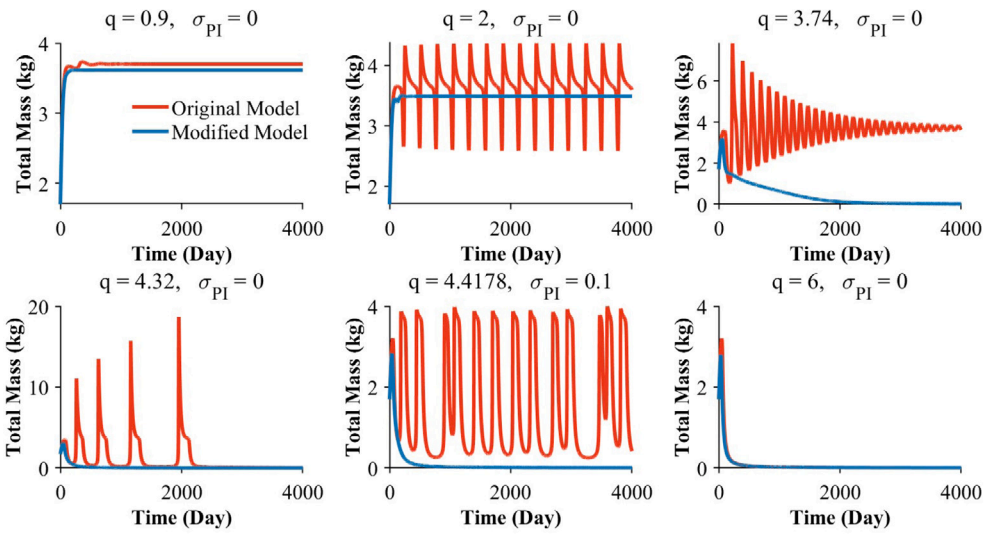


Fig. 2. Dynamical behaviours of total dry matter mass ($M = M_{sh} + M_{rt}$) for the original (red) and modified (blue) Thornley model, Eqs. (1)–(6) and (11)–(16), respectively. Each panel shows the dynamics for different amounts of product inhibition and different transport architectures. All parameters have their values as given in Table 1, except for q and σ_{PI}, which are as marked on the figure.

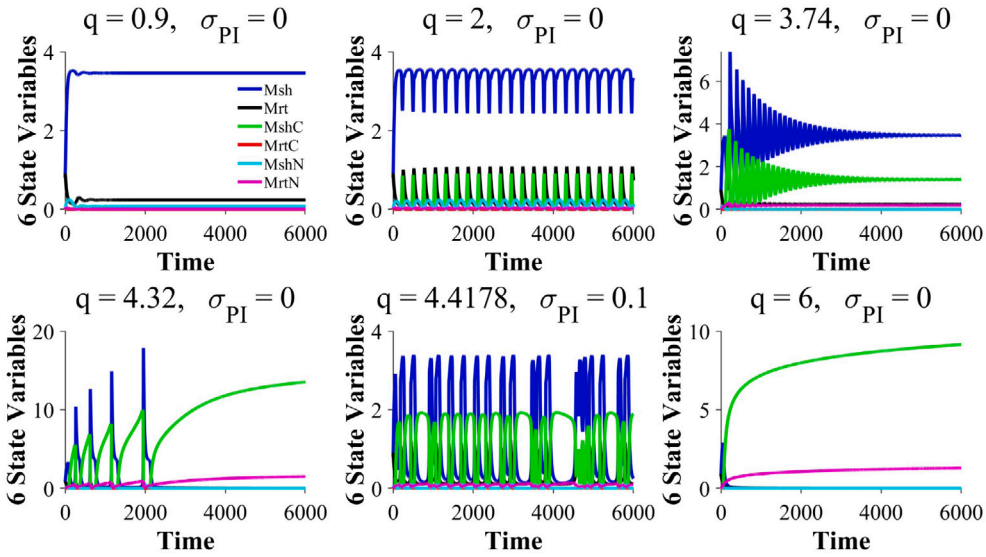


Fig. 3. Different dynamical behaviours of the six state variables in the original Thornley model are included for M_{sh}, M_{rt}, M_{shC}, M_{shN}, M_{rtC}, and M_{rtN} (Eqs. (1) - (6)). All parameters have their values as given in Table 1, except for q and σ_{PI} which were as marked on the figure. In this figure, multiple species behave similarly to each other, in particular, M_{sh}, M_{rt} and M_{shC}. In the bottom left and right, when the plant dies, the root nitrogen and shoot carbon are still produced, increasing to some steady state value, which is not a realistic physical behaviour.

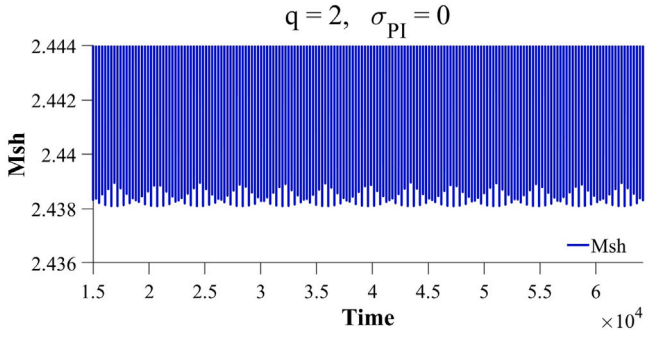


Fig. 4. The detailed dynamics of the shoot dry matter (M_{sh}) for the original Thornley model with $q = 2$ and $\sigma_{PI} = 0$, corresponding to the top middle graph of Fig. 3. Simulating dynamics for a longer period of time, we see a repetitive pattern of subtle variations in oscillation amplitude. .

of time ($t \in [0, 100000]$) to allow it to reach steady state dynamics. We term the early portion, $t \in [0, 10000]$ the initialisation window. We then continue to simulate the model over the period $t \in [100000, 105000]$, that we term the observation window. We examine the dynamics of the plant's total dry matter mass, $M = M_{sh} + M_{rt}$, within both windows and record all peaks of prominence $P > P_{\min} = 0.05$ using MATLAB's `findpeaks` function. This allows us to automatically classify model output into one of the six classes of behaviour visualised in Fig. 2 and 3, using the rules laid out in Table 2.

In Fig. 7, we show maps of how the values of the parameters q and σ_{PI} together affect the dynamics of both the original and modified versions of Thornley's model. The parameters are varied over the ranges $q \in [0, 6]$ and $\sigma_{PI} \in [0, 2]$, which suitably capture the full set of dynamics. For smaller values of q in the original model, we observe the expected steady-state behaviour, but as q increases oscillations begin to appear. Sustained oscillations are associated with higher values of q

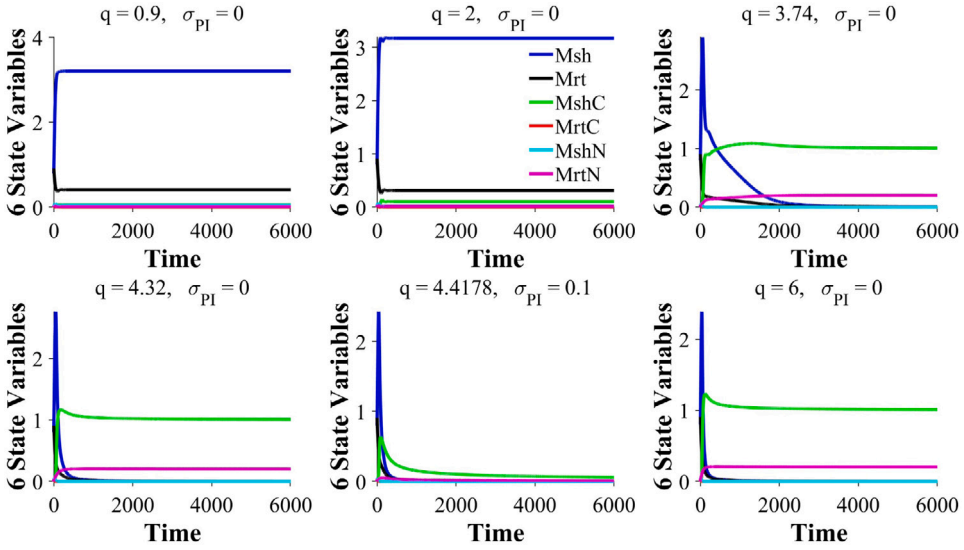


Fig. 5. Different dynamical behaviours of the six state variables in the modified Thornley's model in terms of masses are included: M_{sh}, M_{rt}, M_{shC}, M_{shN}, M_{rtC}, and M_{rtN} (Eqs. (11)–(16)). Nitrogen in the root and carbon in the shoot are still produced when the plant dies, but in small quantities and go to a steady state.

Table 2
The classification system used to automatically identify the dynamics present in a simulated trajectory of the model. Classification uses the dynamics of the total dry matter mass $M = M_{sh} + M_{rt}$ within what we term the initialisation window, $t = [0, 10000]$, and the observation window, $t = [100000, 105000]$. Peaks in the trajectory of M are only recorded if their prominence exceeds $P_{\min} = 0.05$, and $\bar{P}$ refers to the mean prominence of peaks in the relevant window.

Plant Survives: $M > M_{\min}$ anywhere in the observation window	
Steady State	$dM/dt \le M'_{\max}$ within the observation window, and at most one peak during the initialisation window.
Damped Oscillations	$dM/dt \le M'_{\max}$ within the observation window, and more than one peak during the initialisation window.
Sustained Oscillations	$dM/dt > M'_{\max}$ somewhere within the observation window, and all observation window peak prominences P satisfy $\lvert P - \bar{P}\rvert \le \Delta P$.
Period Doubling/Chaos	$dM/dt > M'_{\max}$ somewhere within the observation window, and observation window peak prominences P do not all satisfy $\lvert P - \bar{P}\rvert \le \Delta$.
Plant Dies: $M \le M_{\min}$ during the entire observation window	
Collapse	More than one peak during the initialisation window, and peak heights do not consistently decrease.
Death	One or fewer peaks during the initialisation window, or peak heights consistently decrease.
Parameters: $M_{\min} = 10^{-3}$, $M'_{\max} = 10^{-3}$, $\Delta = 0.01$	

and low values of σ_{PI}, but with $\sigma_{PI} = 0$ there also exists an intermediary window where sustained oscillations are temporarily lost. Further increasing q causes these oscillations to become unstable, potentially passing through a region of period-doubling before entering a regime where the spikes increase in amplitude until the plant collapses and dies. Increasing q further results in the plant dying without this period of oscillation. Increasing σ_{PI} reduces the tendency for oscillation and decreases the critical value of q at which the plant dies, owing to the reduced availability of carbon and nitrogen under increased product inhibition. In particular, for $\sigma_{PI} > 0.2$ and $0 \le q \le 6$, we observe only growth to steady state, possibly approached via damped oscillations, or a dead plant.

In the right panel of Fig. 7, we show how our modified version of the model behaves in response to variations in the same two parameters. Immediately it can be seen that the non-physical behaviours in which plant mass falls into a self-sustained pattern of repeated oscillation, or oscillates with growing amplitude until collapse, no longer occur. In addition to removing any sustained oscillatory dynamics, the modified model also dramatically decreases the proportion of the parameter space that produces temporary oscillations. Additionally, the threshold for plant death shifts in both models. The most fundamental change is when $\sigma_{PI} = 0$ in the original model, the plant dies at $q = 4.7$, but in the modified version, it only survives until $q = 3.7$ (inset). Interestingly, in both models, a small region exists where increasing σ_{PI} allows the plant to survive by receiving fewer nutrients (grey or orange becoming blue upon moving upwards in Fig. 7). Overall, our modification ensures that regardless of the value for the plant architecture parameter q, the plant's predicted behaviours are more biologically reasonable.

Fig. 8 (left column) summarises the dynamics for the two models when q varies and $\sigma_{PI} = 0$, corresponding to the bottom row in Fig. 7. We visualise the total dry matter mass M after running the model long enough to reach steady state, plotting the minimum and maximum values of M to also show the amplitude of any oscillations that persist as long-time behaviour. The top left panel of this figure highlights how sustained oscillations appear for intermediate values of q, and then immediately disappear as q is increased further. We also observe that the mean long-time value for M is fixed until the value of q becomes too large, showing that the sustained oscillations at lower values of q are orbits around the same steady state value of dry matter mass. When sustained oscillations re-appear at around $q = 3.82$, we observe rapidly growing oscillation amplitudes, but the average mass over time decreases until q reaches a high enough value that the plant effectively

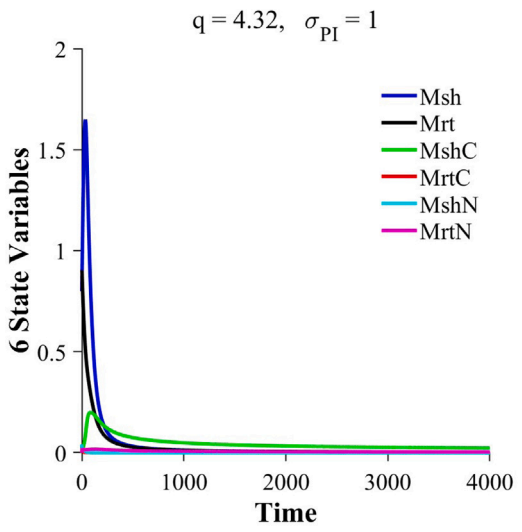


Fig. 6. More realistic dynamical behaviours of the six state variables in the **modified** Thornley model in terms of masses: M_{sh}, M_{rt}, M_{shC}, M_{shN}, M_{rtC}, and M_{rtN} (Eqs. (11) - (16)) when $\sigma_{PI} = 1$. Shoot carbon and root nitrogen decay exponentially towards a steady state extremely near zero, as the plant dies.

dies. In the bottom left panel of Fig. 8, we show the equivalent result for the modified model. Now, the plant consistently reaches a steady state or dies, and the maximum, mean and minimum values for M all coincide due to the absence of sustained oscillations.

Following the importance of the ratio of carbon and nitrogen availability, θ, for setting the plant's state at equilibrium (Section 2.2), we now consider how this balance affects the presence or absence of unrealistic, self-sustained oscillations as q changes. To do so, we vary the four model parameters comprising θ together in a consistent fashion in order to vary its value without introducing other bias. As the base model with parameters as given in Table 1 corresponds to $\theta = 1/4$, we obtain a desired value of $\theta = (k_C f_N)/(k_N f_C)$ by setting the individual parameters to

$$k_C = \kappa\, k_{C0}, \quad k_N = \kappa^{-1}\, k_{N0}, \quad f_C = \kappa^{-1}\, f_{C0}, \quad f_N = \kappa\, f_{N0}, \quad \kappa = (4\theta)^{1/4}\,. \tag{18}$$

Here k_{C0} denotes the base value of k_C given in Table 1, and similarly for the other parameters. We show the effects of such variations in θ on the appearance of sustained oscillations in the right column of Fig. 8.

Across many of the values of θ trialled, we observe the same pattern of sustained oscillations emerging when q becomes sufficiently large, and then plant death as q is increased further. The lower values of θ also display an earlier window of q values where dry matter mass oscillates around its steady state value, as observed already for $\theta = 0.25$. This shows that lower values of q affect how the plant attempts to reach a state of equilibrium, but do not affect this equilibrium itself. This is not surprising, as the parameter q affects only the transport resistance for carbon and nitrogen moving between shoot and, and so its main effects are observed when the dependence of the ability to source nutrients on plant mass becomes too severe when its value is pushed too high. For values $\theta > 1$, the less available nutrient now switches from carbon to nitrogen, but the effect on growth is similar. This is seen in comparing $\theta = 0.5$ and $\theta = 1.5$ in the top right panel of Fig. 8.

Among the different values of θ considered, the plant's ability to reach steady state is least susceptible to the value of q when $\theta = 1$. This is not surprising, as this case corresponds to balance between the plant's production and usage of carbon and nitrogen, and there is consequently less room for the transport term to have a significant effect. However, separate simulations do show that for more extreme values, approximately $q > 6.993$, the plant switches immediately from survival to death. Additionally, we observe that the plant's steady state mass is largest in this case, as both carbon and nitrogen are required for growth and hence efficiency is maximised when both are produced in equal amounts relative to demand.

In the bottom-right panel of Fig. 8, we first see that our modified model continues to avoid sustained oscillations, even as the value of θ is varied. In the modified model, however, changes to q now have a marked effect on the plant's steady state mass, even for comparatively lower values. As in this model there is additional loss of nutrient due to plant litter effects, these results suggest this reduction in pooled carbon and nitrogen makes the plant more sensitive to less severe changes in nutrient transport.

3.3. Steady state eigenvalues

To validate our observations regarding the new models' dynamics, we also examine steady state stability using a linear stability analysis. Linearising the model about the steady state using its Jacobian matrix at that point, a steady state will be stable with respect to small perturbations when all eigenvalues have real part less than zero (Murray, 2002, Appendix B.1). Although we have shown in Section 2.2 a result for the steady state balance between the dry matter mass of shoot and root in the plant, the full details of the model's equilibrium points are not known analytically, let alone the Jacobian eigenvalues that are associated with them. This necessitates a numerical approach.

To numerically determine model equilibrium points in order to examine their stability, we search for the point where the right hand side for all equations is zero using MATLAB's non-linear zero finding algorithm, `fsolve`. We found it necessary to use several techniques to improve the chances of finding and tracking the equilibrium point. Firstly, we use the version of the model defined in terms of concentrations (described in Section 2.3), and further simplify the equation by observing in Eqs. (1) and (2), steady state requires either the corresponding dry matter mass (M_{sh} or M_{rt}) to be zero, or the growth and litter terms to be equal. Seeking a non-death steady state, we assume $M_{sh} \neq 0, M_{rt} \neq 0$ and this allows the growth and litter terms to cancel from the remaining equations (outside of the additional litter term in the modified model). We begin with $q = 0$, which simplifies the transport terms, and then gradually increment q, using the previously found equilibrium point as the initialisation for `fsolve` that finds the next. Where an equilibrium point is not found, we retry with a large number of random initial guesses in `fsolve`. Once the equilibrium point for a given q is found, we evaluate the Jacobian using its closed-form expression, then evaluate its eigenvalues numerically.

We find the eigenvalue behaviour for the baseline model (parameters as in Table 1, corresponding to $\theta = 0.25$) confirms our observations in Section 3.2. For the original model (red curve in Fig. 9, left panel), we find that the real part of one eigenvalue becomes positive, if small, for the window of q that corresponds to sustained oscillations. Linear stability of the equilibrium point is lost, serving to push nearby trajectories onto a limit cycle similar to the one observed in Fig. 3 (top middle panel). As q increases further, linear stability is temporarily regained and then lost again, and does not return. Fig. 7 (left panel, bottom row) shows that a limit cycle returns, until larger q eliminates it and the plant is fated to die.

When product inhibition is switched on in the original model (dashed red line in Fig. 9, top left panel), the largest real part eigenvalue is shifted more negative and the initial window of instability is avoided. When q grows larger, the steady state eventually becomes physically infeasible, in that it becomes required for some substrate nutrient masses to go negative in order for the point to remain an equilibrium point. As we would hope, this switch to an infeasible steady state also results in an eigenvalue with positive real part, such that the non-physical equilibrium point is avoided, and instead the plant dies.

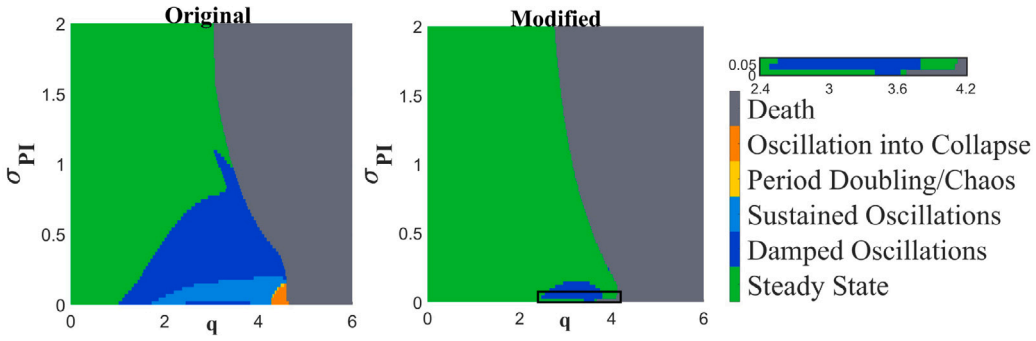


Fig. 7. The effect of the values of parameters q and σ_{PI} on the dynamics of the original (Eqs. (1)–(6)) and the modified (Eqs. (11)–(16)) model. In the original model, many unrealistic behaviours can be seen in the middle region, including several classifications correspond to rapid variations in plant mass (light blue, yellow, orange). Examples of each type of behaviour can be found in Fig. 3. The modified model eliminates these dynamics, and is much less prone to any kind of significant oscillations. Further details are given in the text.

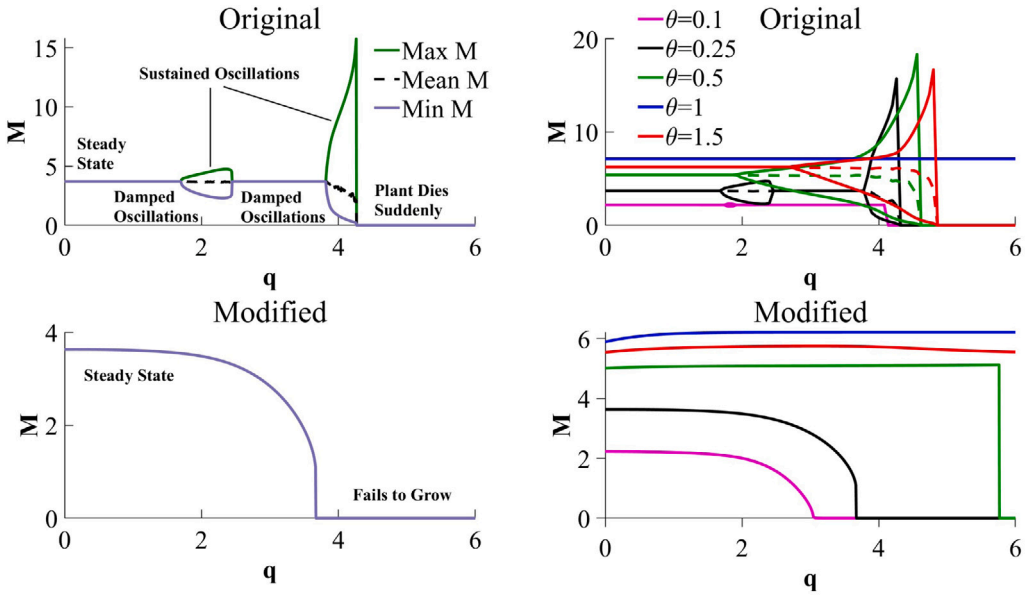


Fig. 8. (Left Column) Emergence of oscillatory and other complex dynamics in the original model, in response to variation in transport parameter q, as indicated by the maximum, mean, and minimum values of parameter M in the long term (q versus $M = M_{sh} + M_{rt}$, the total mass). The results correspond to the bottom row of Fig. 7 ($\theta = 0.25$ and $\sigma_{PI} = 0$). Due to the absence of sustained oscillations, the maximum, mean, and minimum values of dry matter were not plotted in the modified model (corresponds to the right-hand side of Fig. 7). **(Right Column)** Dynamical behaviour of the two models when the parameter θ is varied. In the top right panel (original), for different values of parameter θ, we observe a range of behaviours, including plant death and the possible emergence of oscillations and non-standard oscillations. When $\theta = 1$, we only observe steady state behaviour. In the bottom right panel (Modified model), we observe non-oscillations dynamics for different θ values (Furthur details in text).

We find that, regardless of whether product inhibition is present or not, the stability behaviour of our modified model essentially matches that of the original model with product inhibition turned on. The steady state remains stable until it becomes infeasible, at which point it becomes repelling (blue curves in Fig. 9, top left panel).

In the top right panel of Fig. 9, we display the corresponding results for the case where the plant's carbon and nitrogen availability are in balance, $\theta = 1$, with θ adjusted as in Eq. (7). As previously observed, this case behaves more stably, and all four models are seen to retain all eigenvalue real parts below zero over the range of investigation. Even in this case, however, we see that our modified version of the model serves to shift eigenvalues further into the left half-plane when product inhibition is not present. When product inhibition is present, at least using the values for other parameters given by Thornley (Table 1), we find that both the original and modified versions of the model with $\theta = 1$ show very similar trends in terms of the maximum real part eigenvalue. Trialling other values of θ, we find that with product inhibition off, our modifications to the Thornley model consistently serve to shift the eigenvalue with largest real part leftwards on the complex plane, hence avoiding the unrealistic limit cycle behaviours seen in the original model (see Fig. 9, bottom row). We also find that due to the Thornley model's symmetry between carbon and nitrogen, and the symmetry in how we vary θ through Eq. (18), stability plots akin to those in Fig. 9 are very similar between reciprocal values of θ (that is, $\theta = a$ and $\theta = 1/a$).

Now that we have more rigorously established the benefits of our modified version of the model in terms of its stability, we return to the first-order decay terms proposed as an alternate sink for excess

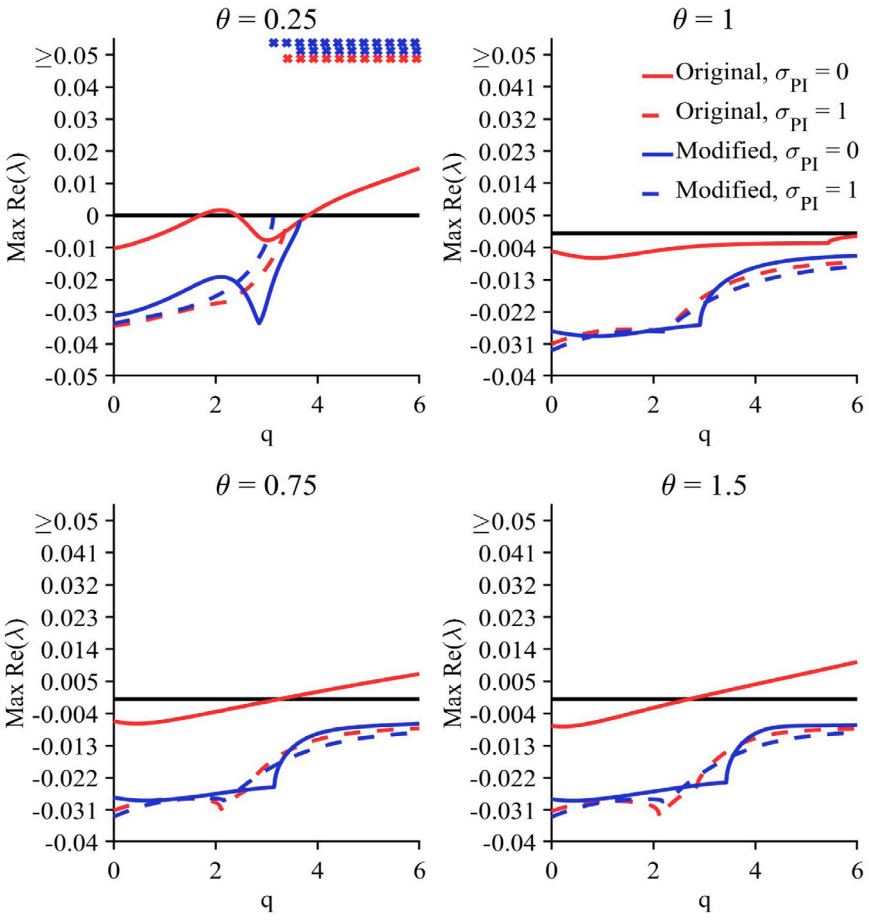


Fig. 9. Stability of the original and modified Thornley model as shown by Jacobian eigenvalues at steady state for different values of θ, with product inhibition on ($\sigma_{PI} = 1$) and off ($\sigma_{PI} = 0$). Non-feasible steady states are marked with ×. The possibility for sustained oscillations in the original model to emerge and disappear as q is increased can be seen in the top left panel, where the maximum real-part eigenvalue for the model without product inhibition briefly crosses into the right side of the complex plane (solid red curve). When carbon and nitrogen demands are in balance ($\theta = 1$, top right panel), there is always a feasible, stable steady state with all eigenvalues having negative real part for this range of q values. Our modified version of the model consistently shifts real part eigenvalues negatively for $\sigma_{PI} = 0$, compared to the original model (solid red curves versus solid blue curves).

nutrients in Section 3. As we detail in Appendix C, the inclusion of these simple type of decay terms can be expected to increase the stability of the model, subtracting their rate constants from diagonal elements of the Jacobian. Unfortunately, this manner of stabilisation means that when these rate constants are too small, unstable, feasible steady states can still exist and produce limit cycle or other unphysical dynamics. Fig. C.1 in the Appendix shows how increasing the decay rate causes the original model's small window of instability around $q = 2$ to eventually disappear, and further increases to the decay rate eventually eliminate the undesirable dynamics that occur for large q values, again by having the steady state switch to being infeasible before the associated largest real-part eigenvalue can move into the right half-plane. Overall, although such sink terms eliminate inappropriate nutrient accumulation and can help with stability, they introduce additional parameters upon which stability directly depends. In contrast, our modification to the model arises directly from a physical argument and with no additional parameters, apparently providing unconditional stability.

4. Discussion and conclusions

Thornley's model provides a detailed description of carbon and nitrogen dynamics in plants that has continued to see use as a component describing allocation of these nutrients in process-based plant models (Higgins et al., 2023). However, the model has arguably not been thoroughly explored mathematically. Here, we have analysed in detail some of the aspects of the original Thornley's model to further understand how it behaves, and which parameters play an important role in controlling its dynamics. In particular, we have identified an important combination of four parameters that represents the balance between carbon and nitrogen, in terms of how the demand due to consumption from growth is met by the rate of nutrient uptake. Moreover, we have identified conditions on some parameters for which the existence of a single feasible steady state in Thornley's model is guaranteed. Importantly, however, we have observed that this steady state is not necessarily linearly stable, and that this can result in unrealistic dynamics including sustained oscillations and even period doubling when the plant architecture parameter q becomes too large.

This led us to explore a modification of Thornley's model that includes additional litter terms, more accurately representing losses of substrate carbon and nitrogen. Our modification is not the only way in which Thornley's model might be changed, but our choice ensures that the loss of plant dry matter due to the litter terms does not cause changes in the concentrations of substrate carbon and nitrogen. Here we have shown that this choice serves to regularise the model against its more inappropriate behaviours, eliminating windows of linear instability and instead ensuring a simple transition into a regime of plant death when the architecture parameter becomes too extreme in value.

Of course, other model modifications in addition to the one we propose might also be beneficial. For example, the way in which the Thornley model establishes steady state might also be looked at. Currently, steady state requires a reduction in carbon/nitrogen uptake as the plant grows, which can be justified by self-shading for carbon (Thornley, 1997) and perhaps crowding effects for nitrogen uptake in roots. A model formulation where the size of a plant at steady state, or some baseline value for such, could be easily set via parameters would be desirable.

Although our modified model is more challenging to analyse from a theoretical perspective, we have performed an extensive sweep over two parameters identified to have significant effects on stability behaviour, and confirmed that the additional litter terms significantly decrease the model's propensity to oscillate towards steady state, and eliminate self-sustaining oscillations. Our parameter sweep also highlights the role played by product inhibition in Thornley's model. Varying the rate of product inhibition of both carbon and nitrogen together by making σ_{PI} a parameter rather than a switch, we find that although increased product inhibition generally serves to regularise the model's behaviour, and triggers death for lower values of the architecture parameter q, there exist small numbers of cases where the opposite occurs. It is particularly notable that decreased access to nutrients can actually rescue a plant, when q is large and product inhibition small. This may indicate a need for a less phenomenological treatment of the effects of plant geometry instead of deferring to an exponent q, or indeed a model that explicitly includes plant structure (Drouet and Pagès, 2007).

Although our modified model behaves more realistically, we still observe that the mass of substrate carbon and nitrogen goes to non-zero steady-state values when the plant dies, particularly when product inhibition is not included. Our analysis of the modified model describes how nutrients behave in the case of plant death and hence explain how product inhibition helps prevent concentrations from growing too fast. We have also considered alternative sink terms for substrate carbon and nitrogen that specifically address the issue of nutrient accumulation in dying plants, an issue that arises because the Thornley model does not impose any hard physical limit on concentration. We find that despite this benefit, and improved stability as compared to the original model, first-order degradation terms do not guarantee stability in the same way our modification does, and also introduce additional parameters into the model. That said, as these terms represent different physical effects, and offer different benefits, the two could always be combined.

Although we have identified a wide range of behaviours, a few unexpected behaviours remain. We have not fully explored how some of the model's other parameters affect these dynamics. We have mathematically identified combinations of parameters that dictate fundamental model behaviours such as the balance of shoot and root mass at steady state, and this may imply a difficulty to identify the values of these parameters individually. We believe it would be an interesting direction of future work to consider parameter identifiability in response to data for this model, and intend to use a Bayesian inference approach to do so in a future publication.

In summary, we have highlighted some unexpected behaviours in the Thornley model of carbon/nitrogen allocation in plants (Thornley, 1997), and have investigated some of the model's key behaviours using both analytical and simulation-driven techniques. By proposing additional litter terms that apply to the substrate carbon and nitrogen in the model, we have put forward a modified model that retains the same mechanistic arguments used to construct Thornley's model, but also achieves a significant reduction in non-physical oscillatory behaviours, and in particular completely eliminates sustained oscillations. Our approach, concentrating on model fundamentals and stress-testing over larger parameter ranges in order to identify possible areas of weakness in a model, is an approach that we believe could be used to benefit other commonly-used mathematical models for plant biology (Bever, 2003). We also suggest that our modification to Thornley's description of carbon/nitrogen allocation be trialled in larger-scale models that use these dynamics as a component (Higgins and Richardson, 2014; Higgins et al., 2023).

CRediT authorship contribution statement

Ati Rostami: Writing – original draft, Software, Investigation, Formal analysis. **Brodie A.J. Lawson:** Writing – review & editing, Supervision, Software, Investigation. **Kevin Burrage:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the ARC Centre of Excellence for Plant Success in Nature and Agriculture, Australia (CE200100015) for financially supporting this research and Prof. Mark Cooper for his constructive feedback and suggestions. We are also grateful to the editor and referees for their constructive comments and support throughout the review process.

Appendix A. Proof of theorems

Theorem 1. *At steady state, the plant's balance of shoot/root is*

$$\frac{\theta M_{sh}}{\left(1+\phi M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)} = \frac{M_{rt}}{\left(1+\phi M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}\frac{M_{rtN}}{M_{rt}}\right)}$$

when no product inhibition ($\sigma_{PI}=0$) is included,

$$M_{rt} = \frac{\theta M_{sh}}{1+\phi M_{sh}(1-\theta)} \tag{A.1}$$

where $\theta = \frac{k_C f_N}{k_N f_C}$ *and* $\phi = \frac{\sigma_{SS}}{K_M}$.

Proof. Equilibria can be found by setting all time derivatives to zero. In the case of (1), (2), (4) and (5) this gives (A.2) to (A.5). Upon substituting (A.2) and (A.3) into (A.4) and (A.5) gives Eqs. (A.2), (A.3), (A.6) and (A.7).

$$\frac{k_G M_{shC} M_{shN}}{M_{sh}} = \sigma_{SS} k_{litt}\left(\frac{M_{sh}^2}{M_{sh}+K_{M,litt}}\right), \tag{A.2}$$

$$\frac{k_G M_{rtC} M_{rtN}}{M_{rt}} = \sigma_{SS} k_{litt}\left(\frac{M_{rt}^2}{M_{rt}+K_{M,litt}}\right), \tag{A.3}$$

$$\frac{1}{x\rho_N}\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right) = f_N k_G \frac{M_{shN}}{M_{sh}} M_{shC}, \tag{A.4}$$

$$\frac{1}{x\rho_C}\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right) = f_C k_G \frac{M_{rtC}}{M_{rt}} M_{rtN}. \tag{A.5}$$

$$\frac{1}{x}\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right) = f_N \rho_N \sigma_{SS} k_{litt}\left(\frac{M_{sh}^2}{M_{sh}+K_{M,litt}}\right), \tag{A.6}$$

$$\frac{1}{x}\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right) = f_C \rho_C \sigma_{SS} k_{litt}\left(\frac{M_{rt}^2}{M_{rt}+K_{M,litt}}\right). \tag{A.7}$$

To further simplify, we label the first terms in Eqs. (3) and (6) as follows

$$F_1 = \frac{M_{sh}}{\left(1+\frac{\sigma_{SS}}{K_M}M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)}, \tag{A.8}$$

$$F_2 = \frac{M_{rt}}{\left(1+\frac{\sigma_{SS}}{K_M}M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}\frac{M_{rtN}}{M_{rt}}\right)}. \tag{A.9}$$

Taking Eqs. (3) and (6) at equilibrium and simplifying by substituting the growth terms from Eqs. (4) and (5), we arrive at the following relations.

$$F_1 k_C f_N = \frac{1}{x}\left[\frac{f_C}{\rho_N}\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right)+\frac{f_N}{\rho_C}\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right)\right], \tag{A.10}$$

$$F_2 k_N f_C = \frac{1}{x}\left[\frac{f_N}{\rho_C}\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right)+\frac{f_C}{\rho_N}\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right)\right]. \tag{A.11}$$

The right-hand sides of both equations are equal, thus we have $k_C f_N F_1 = k_N f_C F_2$, or $\frac{F_2}{F_1} = \frac{k_C f_N}{k_N f_C}$. We denote this ratio θ, so $\theta = \frac{k_C f_N}{k_N f_C}$. After also defining $\phi = \frac{\sigma_{SS}}{K_M}$, we can write the relationship between steady state mass in the shoot and the root as

$$\frac{\theta M_{sh}}{\left(1+\phi M_{sh}\right)\left(1+\frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)} = \frac{M_{rt}}{\left(1+\phi M_{rt}\right)\left(1+\frac{\sigma_{PI}}{J_N}\frac{M_{rtN}}{M_{rt}}\right)}. \tag{A.12}$$

If product inhibition is switched off in the model ($\sigma_{PI} = 0$), then equation (A.12) simplifies to

$$\frac{\theta M_{sh}}{1+\phi M_{sh}} = \frac{M_{rt}}{1+\phi M_{rt}}, \quad \text{or} \quad M_{rt} = \frac{\theta M_{sh}}{1+\phi M_{sh}(1-\theta)}. \ \square \tag{A.13}$$

Theorem 2. *If parameters $\theta < 1$, and $C = \frac{k_C}{\sigma_{SS} k_{litt} f_C} \geq 1$, the existence of single physically feasible steady states in the original Thornley's model is guaranteed.*

Proof. In this scenario, the plant is alive ($M_{sh}, M_{rt} \neq 0$). We analyse equation (A.13), the relationship between masses of shoot matter and root matter at steady state, with the assumption of nonzero masses. We substitute Eqs. (A.6), (A.7), and (A.8) into (A.10). This gives

$$\left(\frac{k_C}{\sigma_{SS} k_{litt} f_C}\right)\left(\frac{M_{sh}}{1+\phi M_{sh}}\right) = \frac{M_{sh}^2}{M_{sh}+K_{M,litt}} + \frac{M_{rt}^2}{M_{rt}+K_{M,litt}}.$$

With $C = \frac{k_C}{\sigma_{SS} k_{litt} f_C}$ and substituting Eq. (A.13), so that after cancelling the nonzero M_{sh} term we have

$$C \times \frac{1}{1+\phi M_{sh}} = \frac{(\theta)^2 M_{sh}}{(1+\phi(1-\theta)M_{sh})^2} \times \frac{1}{\frac{\theta M_{sh}}{1+\phi(1-\theta)M_{sh}} + K_{M,litt}} + \frac{M_{sh}}{M_{sh}+K_{M,litt}}. \tag{A.14}$$

Multiplying through by the denominators in Eq. (A.14) gives a fourth-degree polynomial in M_{sh},

$$\alpha M_{sh}^4 + \beta M_{sh}^3 + \gamma M_{sh}^2 + \delta M_{sh} + \eta = 0, \tag{A.15}$$

where

$$\begin{aligned}
\alpha &= K_{M,litt}[-\phi^3(\theta-1)^2] + \phi^2\theta(\theta-1),\\
\beta &= \phi\theta(C(1-\theta)-2) + K_{M,litt}[\phi^2\left(C(\theta-1)^2-(\theta-1)(\theta-3)\right)],\\
\gamma &= K_{M,litt}[\phi\left(-\theta^2+2\theta-3+C(1-\theta)(2+\theta)\right)] + K_{M,litt}^2[C\phi^2(\theta-1)^2]\\
&\quad + \theta(C-1-\theta),\\
\delta &= K_{M,litt}[C(1+\theta)-(1+\theta^2)] + K_{M,litt}^2[2C\phi(1-\theta)],\\
\eta &= CK_{M,litt}^2.
\end{aligned}$$

We now seek to determine sufficient conditions on the parameters that control how many biologically feasible steady states exist, using Descartes' rule of signs.

In Eq. (A.15), if parameter $\theta < 1$, the coefficient of M_{sh}^4, α, is always negative. The coefficient of M_{sh}^3 is always negative for $C \leq 2/(1-\theta)$ and $\theta < 1$, as β can be written as

$$\beta = K_{M,litt}\phi^2\Big((1-\theta)\big(C(1-\theta)-2-(1-\theta)\big)\Big) + \phi\theta(C(1-\theta)-2). \tag{A.16}$$

Given $\theta < 1$, both bracketed terms are negative so long as $C(1-\theta)-2 \leq 0$, which provides the condition above. The sign of the coefficient of M_{sh}^2 can be positive or negative but will not contribute any additional sign changes if the last two terms are positive. The coefficient of M_{sh}, δ, is positive when $\theta < 1$ so long as

$$C(1+\theta) > 1+\theta^2,$$

and given that $1+\theta > 1+\theta^2$ when $\theta < 1$, this condition is guaranteed to be satisfied so long as $C \geq 1$. The last coefficient, $\eta = CK_{M,litt}^2$, is always positive.

Combining conditions for all coefficients gives $1 < C \leq \frac{2}{1-\theta}$ for one sign change. Due to Descartes' rule of signs, one change of sign in the coefficients dictates that there exists exactly one positive, real root of the polynomial. That is a single physically reasonable steady state value for M_{sh}. So long as the conditions above are satisfied, there is only one steady state, independent of $K_{M,litt}$ and ϕ. Note for Thornley's choice of parameter values as given in Table 1, $C = 4$ and $\theta = 1/4$, satisfying $1 < C \leq \frac{2}{1-\theta}$. $\square$

Theorem 3. *If production (k_C and k_N) versus consumption (f_N and f_C) for carbon and nitrogen are in balance,($\theta = 1$), there is one positive real root in the original Thornley's model.*

Proof. In the case of balance ($\theta = 1$), when production (k_C and k_N) versus consumption (f_N and f_C) for carbon and nitrogen are in balance, Eq. (A.15) reduces to a simpler form,

$$-2\phi M_{sh}^3 + \left(C-2-2K_{M,litt}\phi\right)M_{sh}^2 + 2K_{M,litt}(C-1)M_{sh} + CK_{M,litt}^2 = 0.$$

In the case that the parameter C satisfies $0 < C < 1$, then the coefficients of the first three terms are negative, and the final term is positive. If $C > 1$, we can see only one change of sign in the above equation. If $C \in (1, 2+2K_{M,litt}\phi)$, then the coefficients of the first and second terms are negative, but the coefficients of the third and fourth terms are positive. If parameter $C > 2+2K_{M,litt}\phi$, then the coefficients of the second, third and fourth terms in the given equation are positive. If $C = 1$ or $C = 2+2K_{M,litt}\phi$, then the coefficients of the second or third terms are zero, the fourth term is positive, and the first term is always negative. In these cases (for all $C > 0$), we will always get one change of sign in the coefficients of the polynomial. Descartes' rule of signs thus dictates that there is one positive real root. $\square$

Appendix B. Details about the dynamics of dying plant

Using a non-rigorous limiting argument, we can investigate what happens to the plant when it dies. If we examine Eqs. (3) and (6), we can see that since $M_{sh}, M_{rt} \to 0$, the first terms of these equations are zero. By substituting Eqs. (A.4) and (A.5) instead of the second terms of (3) and (6), respectively, we can obtain the following equations.

$$\frac{dM_{rtN}}{dt} = -\frac{f_N}{f_C}\left(\frac{\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right)}{x\rho_C}\right) - \frac{\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right)}{x\rho_N}, \tag{B.1}$$

$$\frac{dM_{shC}}{dt} = -\frac{f_C}{f_N}\left(\frac{\left(\frac{M_{rtN}}{M_{rt}}-\frac{M_{shN}}{M_{sh}}\right)}{x\rho_N}\right) - \frac{\left(\frac{M_{shC}}{M_{sh}}-\frac{M_{rtC}}{M_{rt}}\right)}{x\rho_C}. \tag{B.2}$$

And it follows that $\frac{dM_{shC}}{dt} - \frac{f_C}{f_N}\frac{dM_{rtN}}{dt} = 0$. Now, by integrating, this becomes

$$M_{shC} = \frac{f_C}{f_N}M_{rtN} + K, \qquad K \text{ is a constant.} \tag{B.3}$$

By using Eq. (B.3) and after simplification then (B.1) becomes

$$\frac{dM_{rtN}}{dt} = -\frac{1}{x}\left(\frac{1}{\rho_N M_{rt}} + \frac{1}{\rho_C M_{sh}}\right) M_{rtN} + \frac{1}{x}\left(\frac{f_N M_{rtC}}{f_C \rho_C M_{rt}} + \frac{M_{shN}}{\rho_N M_{sh}} - \frac{K f_N}{f_C \rho_C M_{sh}}\right). \quad \text{(B.4)}$$

For appropriate ranges of q and with M_{sh}, M_{rt}, M_{shN}, $M_{rtC} \to 0$ then the solution of Eq. (B.4) is an exponential approach to nonzero steady state value. This is obviously a non-physical phenomenon. These dynamics can also be observed in simulations of the model where the plant functionally dies (Section 3).

Appendix C. Investigation of the linear decay model

Owing to the accumulation of substrate carbon and nitrogen in dying plants, observed for example in Fig. 5, there is a motivation to introduce sink terms for substrate carbon and nitrogen that allow them to be removed even when the plant is dying and has very small mass. However, if we replace our modification to Thornley's model with an alternative sink term for substrate nutrients, it is important that we examine the effect of such a term on the model's stability.

Here, we consider the simple case where a first-order decay term is added for substrate nutrients, i.e. to Eqs. (3)–(6) or (13)–(16). As an example, Eq. (3) becomes

$$\frac{dM_{shC}}{dt} = \frac{k_C M_{sh}}{\left(1 + \frac{\sigma_{SS}}{K_M} M_{sh}\right)\left(1 + \frac{\sigma_{PI}}{J_C}\frac{M_{shC}}{M_{sh}}\right)} - f_C k_G \frac{M_{shC}}{M_{sh}} M_{shN} - \frac{1}{x\rho_C}\left(\frac{M_{shC}}{M_{sh}} - \frac{M_{rtC}}{M_{rt}}\right) - \delta_C M_{shC},$$

where δ_C with units of d^{-1} is the decay rate. A separate decay rate could be specified for nitrogen, δ_N, or indeed rates could also differ between shoot and root. Here, however, we simply set $\delta_C = \delta_N = \delta$.

For these choices of sink term and parameterisation, we have that the Jacobian matrix is of the form

$$\mathbf{J} = \mathbf{J}_0 - \delta \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

Where $\mathbf{J}_0$ is the Jacobian matrix evaluated without inclusion of the decay terms. Subtraction of a matrix close to $\delta\mathbf{I}$ with $\mathbf{I}$ the identity matrix suggests that the eigenvalues will approximately shift leftwards in the complex plane by δ. This result could be formalised somewhat by the Gershgorin circle theorem, although we observed via numerical evaluations that one circle's radius is sufficient to engulf the others, and this theorem brings little information. Nevertheless, we find that increasing δ indeed shifts at least the eigenvalue important for stability leftward in the complex plane (down on Fig. C.1). Hence $\delta > 0$ confers improved model stability, and larger values of δ are more protective than smaller values, shifting the eigenvalue responsible for stability further away from the complex axis and hence helping its real part remain negative as $\mathbf{J}_0$ changes.

In Fig. C.1, we show the stability plots for different values of δ, using the original model with no product inhibition. As already observed in the main document, the base model shows a window of instability around $q = 2$, and then a return to instability when q becomes too large. By introducing first-order decay, the real part of the stability-defining eigenvalue decreases as expected, and when δ is increased sufficiently, the eigenvalue inside of the first instability window is shifted negative enough to remove instability from the model. For the larger values of q, $\delta = 0.01$ is not sufficiently large enough to avoid positive eigenvalues associated with a feasible steady state, and hence undesirable behaviour persists until δ is driven even higher. Values of δ required to obtain stability in different scenarios will be *a priori* unknown.

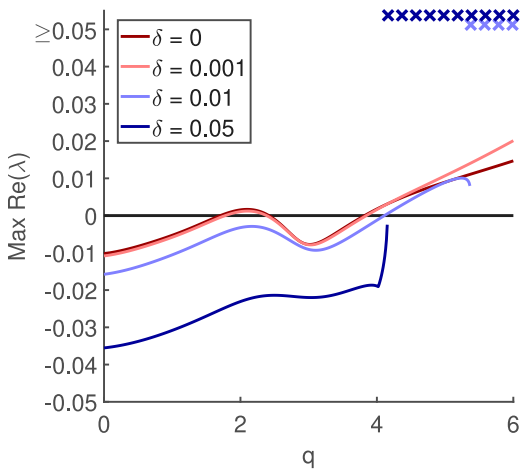


Fig. C.1. Effects on stability of the original Thornley model, without product inhibition, for different decay rates. As in the main document, cross marks (×) indicate a non-feasible steady state. Sufficiently increasing decay rate δ avoids the window of instability, and further increasing δ prevents eigenvalues from ever becoming positive before the jump to an infeasible, unstable steady state (and hence plant death).

References

Alligood, K.T., Sauer, T.D., Yorke, J.A., Chillingworth, D., 1996. Chaos: An Introduction to Dynamical Systems. Springer-Verlag, New York.

Ban, B., Lee, M., Ryu, D., 2019. ODE network model for nonlinear and complex agricultural nutrient solution system. In: 2019 International Conference on Information and Communication Technology Convergence. ICTC, pp. 996–1001. http://dx.doi.org/10.1109/ICTC46691.2019.8939946.

Bever, J.D., 2003. Soil community feedback and the coexistence of competitors: conceptual frameworks and empirical tests. New Phytol. 157 (3), 465–473. http://dx.doi.org/10.1046/j.1469-8137.2003.00714.x.

Chew, Y.H., Smith, R.W., Jones, H.J., Seaton, D.D., Grima, R., Halliday, K.J., 2014. Mathematical models light up plant signaling. Plant Cell 26 (1), 5–20. http://dx.doi.org/10.1105/tpc.113.120006.

Dewar, R.C., 1993. A root-shoot partitioning model based on carbon-nitrogen-water interactions and munch phloem flow. Funct. Ecol. 7 (3), 356–368, https://www.jstor.org/stable/2390216.

Dewar, R.C., Medlyn, B.E., Mcmurtrie, R.E., 2002. A mechanistic analysis of light and carbon use efficiencies. Plant, Cell Environ. 21 (6), 573–588. http://dx.doi.org/10.1046/j.1365-3040.1998.00311.x.

Dorneles, M., Da Silva, C., Gomes, A., 2001. A model for hyphae effects in phosphorus absorption by plants. Ecol. Model. 142 (1–2), 83–89. http://dx.doi.org/10.1016/S0304-3800(01)00280-0.

Drouet, J.-L., Pagès, L., 2007. GRAAL-CN: A model of GRowth, architecture and ALlocation for carbon and nitrogen dynamics within whole plants formalised at the organ level. Ecol. Model. 206 (3), 231–249. http://dx.doi.org/10.1016/j.ecolmodel.2007.03.036, URL https://www.sciencedirect.com/science/article/pii/S0304380007001810.

Dun, E.A., Hanan, J., Beveridge, C.A., 2009. Computational modeling and molecular physiology experiments reveal new insights into shoot branching in pea. Plant Cell 21 (11), 3459–3472. http://dx.doi.org/10.1105/tpc.109.069013.

Farquhar, G.D., von Caemmerer, S.v., Berry, J.A., 1980. A biochemical model of photosynthetic CO 2 assimilation in leaves of c 3 species. Planta 149 (1), 78–90. http://dx.doi.org/10.1007/BF00386231.

Feller, C., Favre, P., Janka, A., Zeeman, S.C., Gabriel, J.-P., Reinhardt, D., 2015. Mathematical modeling of the dynamics of shoot-root interactions and resource partitioning in plant growth. PloS One 10 (7), e0127905. http://dx.doi.org/10.1371/journal.pone.0127905.

Higgins, S.I., Conradi, T., Muhoko, E., 2023. Shifts in vegetation activity of terrestrial ecosystems attributable to climate trends. Nat. Geosci. 16 (2), 147–153. http://dx.doi.org/10.1038/s41561-022-01114-x.

Higgins, S.I., O'Hara, R.B., Bykova, O., Cramer, M.D., Chuine, I., Gerstner, E.-M., Hickler, T., Morin, X., Kearney, M.R., Midgley, G.F., et al., 2012. A physiological analogy of the niche for projecting the potential distribution of plants. J. Biogeogr. 39 (12), 2132–2145. http://dx.doi.org/10.1111/j.1365-2699.2012.02752.x.

Higgins, S.I., Richardson, D.M., 2014. Invasive plants have broader physiological niches. Proc. Natl. Acad. Sci. 111 (29), 10610–10614. http://dx.doi.org/10.1073/pnas.140607511.

Krämer, K., Brock, J., Heyer, A.G., 2022. Interaction of nitrate assimilation and photorespiration at elevated CO2. Front. Plant Sci. 13, 897924. http://dx.doi.org/10.3389/fpls.2022.897924.

Küstner, L., Nägele, T., Heyer, A.G., 2019. Mathematical modeling of diurnal patterns of carbon allocation to shoot and root in arabidopsis thaliana. Npj Syst. Biology Appl. 5 (1), 4, https://www.nature.com/articles/s41540-018-0080-1.

Lacointe, A., 2000. Carbon allocation among tree organs: a review of basic processes and representation in functional-structural tree models. Ann. For. Sci. 57 (5), 521–533. http://dx.doi.org/10.1051/forest:2000139.

Linker, R., Johnson-Rutzke, C., 2005. Modeling the effect of abrupt changes in nitrogen availability on lettuce growth, root–shoot partitioning and nitrate concentration. Agricult. Sys. 86 (2), 166–189. http://dx.doi.org/10.1016/j.agsy.2004.08.008, URL https://www.sciencedirect.com/science/article/pii/S0308521X04001751.

Lunn, J.E., 2007. Compartmentation in plant metabolism. J. Exp. Bot. 58 (1), 35–47. http://dx.doi.org/10.1093/jxb/erl134.

Mauseth, J.D., 2014. Botany: An Introduction to Plant Biology. Jones and Bartlett Publishers, ISBN: 1284157350, https://ebookcentral.proquest.com/lib/qut/reader.action?docID=59055622.

McCown, R., Hammer, G., Hargreaves, J., Holzworth, D., Freebairn, D., 1996. APSIM: a novel software system for model development, model testing and simulation in agricultural systems research. Agricult. Sys. (ISSN: 0308-521X) 50 (3), 255–271. http://dx.doi.org/10.1016/0308-521X(94)00055-V, URL https://www.sciencedirect.com/science/article/pii/0308521X9400055V.

Monsalve-Bravo, G.M., Lawson, B.A., Drovandi, C., Burrage, K., Brown, K.S., Baker, C.M., Vollert, S.A., Mengersen, K., McDonald-Madden, E., Adams, M.P., 2022. Analysis of sloppiness in model simulations: Unveiling parameter uncertainty when mathematical models are fitted to data. Sci. Adv. 8 (38), eabm5952. http://dx.doi.org/10.1126/sciadv.abm5952.

Murray, J.D., 2002. Mathematical Biology I: An Introduction, third ed. Springer-Verlag, New York.

Payvandi, S., Daly, K.R., Zygalakis, K., Roose, T., 2014. Mathematical modelling of the phloem: The importance of diffusion on sugar transport at osmotic equilibrium. Bull. Math. Biol. 76, 2834–2865. http://dx.doi.org/10.1007/s11538-014-0035-7.

Pitchford, J.W., Avello, P., 2022. ODE (ordinary differential equation) models for plant circadian networks: What the models are and how they should be used. Plant Circadian Netw.: Methods Protoc. 2398, 75–88. http://dx.doi.org/10.1007/978-1-0716-1912-4_7.

Polynikis, A., Hogan, S., Di Bernardo, M., 2009. Comparing different ODE modelling approaches for gene regulatory networks. J. Theoret. Biol. 261 (4), 511–530. http://dx.doi.org/10.1016/j.jtbi.2009.07.040.

Potkay, A., Hölttä, T., Trugman, A.T., Fan, Y., 2022. Turgor-limited predictions of tree growth, height and metabolic scaling over tree lifespans. Tree Physiol. 42 (2), 229–252. http://dx.doi.org/10.1093/treephys/tpab094.

Rohde, W., Forni, F., 2023. Lettuce modelling for growth control in precision agriculture. Eur. J. Control (ISSN: 0947-3580) 74, 100843. http://dx.doi.org/10.1016/j.ejcon.2023.100843, URL https://www.sciencedirect.com/science/article/pii/S0947358023000729, 2023 European Control Conference Special Issue.

Seginer, I., 2003. A dynamic model for nitrogen-stressed lettuce. Ann. Botany 91 (6), 623–635. http://dx.doi.org/10.1093/aob/mcg069.

Seki, M., Feugier, F.G., Song, X.-J., Ashikari, M., Nakamura, H., Ishiyama, K., Yamaya, T., Inari-Ikeda, M., Kitano, H., Satake, A., 2015. A mathematical model of phloem sucrose transport as a new tool for designing rice panicle structure for high grain yield. Plant Cell Physiol. 56 (4), 605–619. http://dx.doi.org/10.1093/pcp/pcu191.

Sievänen, R., Nikinmaa, E., Nygren, P., Ozier-Lafontaine, H., Perttunen, J., Hakula, H., 2000. Components of functional-structural tree models. Ann. For. Sci. 57 (5), 399–412, https://www.afs-journal.org/articles/forest/pdf/2000/05/f0501.pdf.

Thornley, J.H.M., 1972. A balanced quantitative model for root: Shoot ratios in vegetative plants. Ann. Botany 36 (145), 431–441, http://www.jstor.org/stable/42753756.

Thornley, J.H.M., 1991. A transport-resistance model of forest growth and partitioning. Ann. Botany 68 (3), 211–226. http://dx.doi.org/10.1093/oxfordjournals.aob.a088246.

Thornley, J.H.M., 1995. Shoot: Root allocation with respect to c, n and p: an investigation and comparison of resistance and teleonomic models. Ann. Botany 75 (4), 391–405, http://www.jstor.org/stable/42761731.

Thornley, J.H.M., 1997. Modelling allocation with transport/conversion processes. Silva Fenn. 31 (3), 341–355, http://hdl.handle.net/1975/8532.

Thornley, J.H.M., 1998a. Modelling shoot:Root relations: the only way forward? Ann. Botany 81 (2), 165–171. http://dx.doi.org/10.1006/anbo.1997.0529.

Thornley, J.H.M., 1998b. Dynamic model of leaf photosynthesis with acclimation to light and nitrogen. Ann. Botany 831 (3), 421–430. http://dx.doi.org/10.1006/anbo.1997.0575.

Thornley, J.H.M., 1999. Modelling stem height and diameter growth in plants. Ann. Botany 84 (2), 195–205. http://dx.doi.org/10.1006/anbo.1999.0908.

Thornley, J.H.M., France, J., 2007. Mathematical Models in Agriculture: Quantitative Methods for the Plant, Animal and Ecological Sciences. CABI, Wallingford UK, pp. 385–449.

Thornley, J.H.M., Parsons, A.J., 2014. Allocation of new growth between shoot, root and mycorrhiza in relation to carbon, nitrogen and phosphate supply: teleonomy with maximum growth rate. J. Theoret. Biol. 342, 1–14. http://dx.doi.org/10.1016/j.jtbi.2013.10.003.

Weisberg, P.J., Bonavia, F., Bugmann, H., 2005. Modeling the interacting effects of browsing and shading on mountain forest tree regeneration (picea babies). Ecol. Model. 185 (2–4), 213–230. http://dx.doi.org/10.1016/j.ecolmodel.2004.12.004.

Zaffaroni, M., Cunniffe, N.J., Bevacqua, D., 2020. An ecophysiological model of plant–pest interactions: the role of nutrient and water availability. J. R. Soc. Interface 17 (172), 20200356. http://dx.doi.org/10.1098/rsif.2020.0356.